

Jatin Nath

New Delhi, Delhi, India

📞 +91 96430 07622 — ✉️ jatinnath1111@gmail.com — Portfolio: jatin.portfolio.app
linkedin.com/in/jatinnath1 — github.com/jatinnathh

Summary

Computer Science (AI & Data Science) undergraduate seeking a **Software Engineering Intern** or entry-level SDE role. Strong foundations in **Data Structures & Algorithms**, **Object-Oriented Programming**, **system design**, and **backend engineering**. **Hackathon winner (TetherX, VIT Chennai)** for building a real-time hospital workflow automation platform. Experienced in building and deploying scalable, Dockerized applications with **REST APIs**, **FastAPI**, and **Node.js/Express** following **Agile** development practices. Built **NeuroFusion**, a production-grade Dockerized ML application, and **ModelForge**, an AutoML platform deployed on **AWS**. Proficient in **Python**, **C++**, **JavaScript**, **SQL**, with hands-on experience in **CI/CD workflows**, **unit testing**, **code review**, and version control (Git).

Experience

Innova Solutions

Software Engineering Intern

On-site

May 26, 2025 – July 5, 2025

- **Links:** GitHub: <https://github.com/jatinnathh/NeuroFusion>
- Engineered **NeuroFusion**, a full-stack text-to-image application; resolved **20+** critical bugs through systematic debugging and **unit testing**, improving stability and user experience across web and mobile.
- Designed and implemented a **RESTful backend** with **RBAC**, relational data modeling, and **MySQL**-backed storage supporting multi-user access control workflows.
- Built an asynchronous **prompt queuing system** handling **15+ concurrent requests** via sequential GPU execution using **microservices** principles for reliability and throughput.
- Optimized GPU scheduling logic, achieving a **40% reduction in GPU idle time** — demonstrating performance profiling and systems-level thinking.
- Containerized the application using **Docker**, set up **CI/CD pipelines** with **GitHub Actions**, and managed source control with **Git**, ensuring reproducible builds and streamlined team collaboration.

Education

Indian Institute of Information Technology

B.Tech, Computer Science (AI and Data Science)

Kottayam, India

2023 – 2027 (Expected)

Achievements

2026: Winner (1st Place) — TetherX Hackathon, VIT Chennai — Built *MediFlow*, a real-time hospital inter-department workflow automation system featuring dual-auth RBAC, WebSocket chat, AI diagnostics, and automated billing.

Projects

MediFlow — Hospital Inter-Department Workflow Automation (Winner, TetherX Hackathon — VIT Chennai) 2026
Stack: Next.js, TypeScript, PostgreSQL, Prisma ORM, Socket.IO, JWT, Hugging Face, Google Gemini AI, Razorpay, Vercel

- Website: <https://turbo-s-final.vercel.app/> — GitHub: <https://github.com/jatinnathh/TurboS-Final>
- Built a **full-stack Next.js/TypeScript** platform automating patient workflows across hospital departments with **role-based access control** (dual-auth: Clerk + JWT), **REST API** routes, and **PostgreSQL** (NeonDB) via **Prisma ORM**.
- Engineered a **real-time communication layer** using a standalone **Socket.IO WebSocket server** with shared JWT authentication, enabling live doctor-patient chat, typing indicators, and push notifications.
- Designed an **inter-department workflow engine** supporting referrals, ward transfers, and lab test ordering (PENDING → IN_PROGRESS → COMPLETED) — each action tracked with timestamped audit logs and a live activity feed.
- Integrated **Hugging Face ML models** for skin cancer image classification, **Google Gemini AI** for clinical decision support, and **Razorpay** for automated billing and payment processing.

Stack: Python, Streamlit, scikit-learn, AWS EC2, REST API, Git, OOP

- Website: <http://51.21.222.225:8501/> — GitHub: <https://github.com/jatinnathh/MLGenie>
- Designed and deployed a **full-stack AutoML platform** on **AWS EC2** using **Python**, **Streamlit**, and **scikit-learn**, with modular OOP architecture for extensibility.
- Implemented **RESTful API** endpoints for model training, evaluation, and retrieval; automated data preprocessing and feature analysis pipelines, reducing manual effort by **~70%**.
- Applied **software engineering principles** (separation of concerns, DRY, SOLID) and **Agile** practices to produce a maintainable, production-ready codebase managed with **Git** and peer **code review**.

Certifications

Caltech

Post Graduate Program in AI and Machine Learning

2024

- Project-based curriculum covering AI, ML, Deep Learning, Data Science, and Generative AI.
- Capstone guided by Dr. Rick Hefner; certificate endorsed by Caltech CTME.

IBM

IBM Data Science Professional Certificate

2024

- Mastered Python for data manipulation, analysis, and visualization. Issued: June 5, 2024.

Skills

Languages: Python, C++, JavaScript, TypeScript, SQL, C

CS Fundamentals: Data Structures & Algorithms, OOP, System Design, Operating Systems, DBMS, Computer Networks, Software Engineering Principles

Backend & APIs: Node.js, Express.js, FastAPI, Flask, REST API Design, Microservices, Socket.IO, MongoDB, MySQL, PostgreSQL, Prisma, Redis

DevOps & Tools: Docker, CI/CD (GitHub Actions), AWS (EC2, S3), Git, GitHub, Linux, Vercel, MLflow, Streamlit, Unit Testing, Agile/Scrum

Frontend: React, Next.js, TypeScript, Tailwind CSS

ML/AI Frameworks: PyTorch, TensorFlow, scikit-learn, OpenCV, Hugging Face, Pandas, NumPy

Coursework

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| ○ Data Structures and Algorithms | ○ Database Management Systems |
| ○ Operating Systems | ○ Object-Oriented Programming |
| ○ Probability, Statistics & Random Processes | ○ Discrete Mathematics |
| ○ Calculus and Linear Algebra | ○ Computer Networks |

Additional

Languages: English (Fluent), Hindi (Fluent)